



**INNOVATION. AUTOMATION. ANALYTICS**

# **PROJECT ON** **World Wide Energy Consumption**

DONE BY:  
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# About me



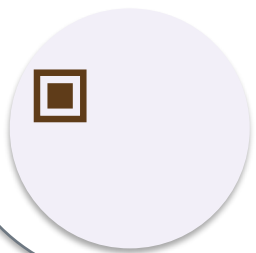
- **Background? (B-tech or M-tech)**

Btech



- **Why you want to learn Data Science**

I want to learn Data Science to build intelligent solutions and solve real-world problems using data.



- **Any work experience**

I am a fresher and do not have any work experience yet.

# Objective of the Project

- Analyze energy consumption, production and emissions across countries
- Compare energy data with GDP and population
- Perform country-wise and year-wise analysis using SQL
- Calculate per-capita and ratio-based metrics
- Identify trends and meaningful insights from the data

# Database Context

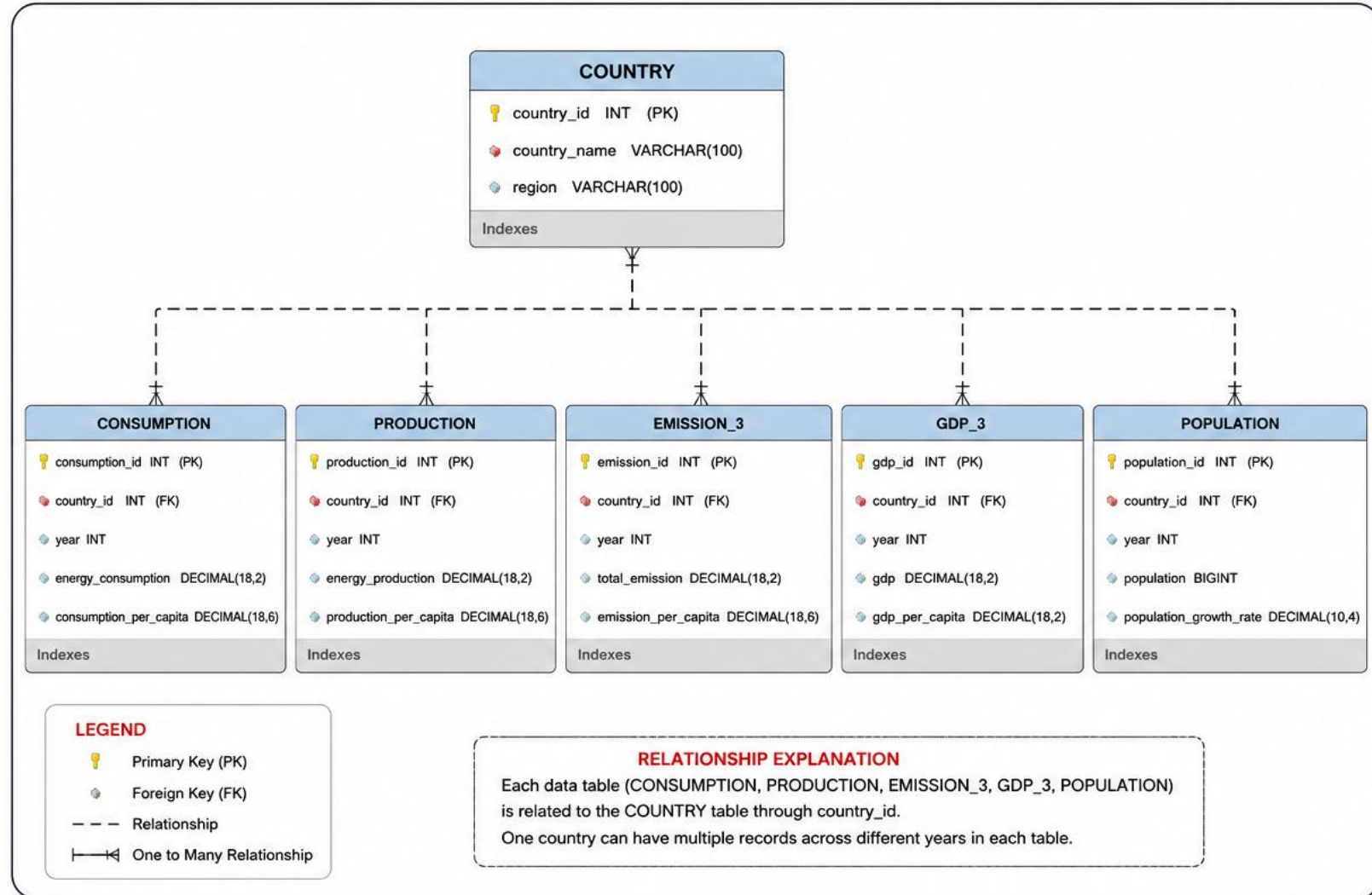
- **COUNTRY** – Central reference table
- **CONSUMPTION** – Energy consumption data
- **PRODUCTION** – Energy production data
- **EMISSION\_3** – Emission data
- **GDP\_3** – GDP data
- **POPULATION** – Population data

# ERR DIAGRAM

## Topic :

## ENERGY CONSUMPTION ANALYSIS

This ER diagram represents the structure of the Energy Consumption database used for analyzing global energy consumption, production, emissions, GDP and population trends across countries and years.



# Q&A Slide

## Total Emission by Country :

Q. What is the total emission per country for the most recent year available?

Query :

```
SELECT country, SUM(emission) AS total_emission
FROM emission_3
WHERE year = (SELECT MAX(year) FROM emission_3)
GROUP BY country
ORDER BY total_emission DESC;
```

Key Insights :

- The most recent year in the emission dataset is 2023.
- Countries are ranked by total emissions after aggregating all energy types.

Output :

| Country       | Emission |
|---------------|----------|
| China         | 24,391   |
| United States | 9,591    |
| India         | 5,641    |
| Russia        | 3,689    |
| Japan         | 1,920    |

# Q&A Slide

## Top 5 Countries by GDP :

Q. What are the top 5 countries by GDP in the most recent year?

### Query :

```
SELECT Country, Value AS GDP
FROM gdp_3
WHERE year = (SELECT MAX(year) FROM gdp_3)
ORDER BY GDP DESC
LIMIT 5;
```

### Output :

| Country       | Value  |
|---------------|--------|
| China         | 28,673 |
| United States | 22,679 |
| India         | 11,660 |
| Japan         | 5,180  |
| Germany       | 4,464  |

### Key Insights :

- The latest GDP year in the dataset is 2024.
- The output shows the five countries with the highest GDP values.

# Q&A Slide

## Energy Production vs Consumption :

Q. Compare energy production and consumption by country and year.

### Query :

```
SELECT p.country, p.energy, p.year,
       p.production, c.consumption
FROM production p
JOIN consumption c
  ON p.country = c.country
 AND p.year = c.year
 AND p.energy = c.energy;
```

### Key Insights :

- Production and consumption are matched using country, year and energy.
- This makes it possible to compare supply and usage for the same energy type.

### Output :

| Country     | Year  | Energy                                    | Production | Consumption |
|-------------|-------|-------------------------------------------|------------|-------------|
| Afghanistan | 2,020 | Coal (quad Btu)                           | 0.035      | 0.021       |
| Afghanistan | 2,020 | Natural gas (quad Btu)                    | 0.003      | 0.003       |
| Afghanistan | 2,020 | Nuclear (quad Btu)                        | 0          | 0           |
| Afghanistan | 2,020 | Nuclear, renewables, and other (quad Btu) | 0.004      | 0.021       |
| Afghanistan | 2,020 | Petroleum and other liquids (quad Btu)    | 0          | 0.115       |
| Afghanistan | 2,020 | Renewables and other (quad Btu)           | 0.004      | 0.021       |



# Q&A Slide

## Population Growth vs Total Emissions :

Q. How has population growth affected total emissions in each country?

### Query :

```
SELECT p.countries AS country, p.year,
       p.Value AS population,
       SUM(e.emission) AS total_emission
FROM population p
JOIN emission_3 e
  ON p.countries = e.country
 AND p.year = e.year
GROUP BY p.countries, p.year, p.Value
ORDER BY p.countries, p.year;
```

### Output :

| Countries     | Year  | Value     | Emission |
|---------------|-------|-----------|----------|
| China         | 2,023 | 1,422,585 | 24,391   |
| United States | 2,023 | 336,445   | 9,591    |
| India         | 2,023 | 1,438,070 | 5,641    |
| Russia        | 2,023 | 145,440   | 3,689    |
| Japan         | 2,023 | 124,371   | 1,920    |

### Key Insights :

- Population and total emissions are combined using country and year.
- The result supports comparison of demographic size and emission levels over time.

# Q&A Slide

## Average Yearly Change in Per-Capita Emissions :

Q. What is the average yearly change in emissions per capita for each country?

### Query :

```
WITH yearly_pc AS (  
  SELECT country, year,  
         AVG(`per capita emission`) AS pc_emission  
  FROM emission_3  
  GROUP BY country, year  
)  
changes AS (  
  SELECT country, year,  
         pc_emission -  
         LAG(pc_emission) OVER  
           (PARTITION BY country ORDER BY year) AS yearly_change  
  FROM yearly_pc  
)  
SELECT country, AVG(yearly_change) AS avg_yearly_change  
FROM changes
```

### Key Insights :

- Per-capita emissions are compared with the previous year using LAG().
- The final average summarizes the yearly change for each country.

### Output :

| Country             | Yearly Change |
|---------------------|---------------|
| Albania             | 0             |
| Algeria             | 0             |
| American Samoa      | 0             |
| Angola              | 0             |
| Antigua and Barbuda | 0             |

# Q&A Slide

## Emission-to-GDP Ratio :

Q. What is the emission-to-GDP ratio for each country by year?

### Query :

```
SELECT e.country, e.year,
       SUM(e.emission) / g.Value
       AS emission_to_gdp_ratio
FROM emission_3 e
JOIN gdp_3 g
  ON e.country = g.Country
 AND e.year = g.year
GROUP BY e.country, e.year, g.Value
ORDER BY e.country, e.year;
```

### Output :

| Country              | Year  | Emission Gdp Ratio |
|----------------------|-------|--------------------|
| Netherlands Antilles | 2,023 | 7.811              |
| North Korea          | 2,023 | 2.944              |
| U.S. Virgin Islands  | 2,023 | 2.524              |
| Turkmenistan         | 2,023 | 2.077              |
| Trinidad and Tobago  | 2,023 | 2.005              |

### Key Insights :

- The ratio relates total emissions to GDP for the same country and year.
- Lower values can indicate lower emissions relative to economic output.

# Q&A Slide

## Global Average GDP, Emission & Population :

Q. What is the global average GDP, emission, and population by year?

### Query :

```
SELECT g.year,
       AVG(g.Value) AS avg_gdp,
       AVG(e.total_emission) AS avg_emission,
       AVG(p.Value) AS avg_population
FROM gdp_3 g
JOIN (
  SELECT country, year, SUM(emission) AS total_emission
  FROM emission_3
  GROUP BY country, year
) e ON g.Country = e.country AND g.year = e.year
JOIN population p
  ON g.Country = p.countries AND g.year = p.year
GROUP BY g.year
ORDER BY g.year;
```

### Output :

| Year  | Avg Gdp | Avg Emission | Avg Population |
|-------|---------|--------------|----------------|
| 2,020 | 624.401 | 339.017      | 39,363         |
| 2,021 | 665.349 | 354.708      | 39,691         |
| 2,022 | 688.699 | 362.021      | 40,048         |
| 2,023 | 711.818 | 370.520      | 40,411         |

### Key Insights :

- GDP, total emissions and population are averaged year by year.
- This provides a global-level summary of economic, environmental and demographic trends.

## Key Findings & Highlights

|                                                                                     |                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <ul style="list-style-type: none"><li>• <b>Emission analysis</b><br/>Identified countries with the highest total emissions in the most recent year available.</li></ul>                                                                                                      |
|    | <ul style="list-style-type: none"><li>• <b>GDP comparison</b><br/>Analyzed the top 5 countries by GDP to understand major economies.</li></ul>                                                                                                                               |
|    | <ul style="list-style-type: none"><li>• <b>Energy production vs consumption</b><br/>Compared energy production and consumption across countries, years, and energy types.</li></ul>                                                                                          |
|    | <ul style="list-style-type: none"><li>• <b>Year-over-year emission trends</b><br/>Analyzed whether global emissions increased or decreased over time.</li></ul>                                                                                                              |
|    | <ul style="list-style-type: none"><li>• <b>Population impact</b><br/>Compared population growth with total emissions to understand their relationship.</li></ul>                                                                                                             |
|    | <ul style="list-style-type: none"><li>• <b>Per-capita emissions</b><br/>Measured the average yearly change in emissions per capita for each country.</li></ul>                                                                                                               |
|   | <ul style="list-style-type: none"><li>• <b>Emission efficiency</b><br/>Calculated the emission-to-GDP ratio to compare environmental impact relative to economic output.</li></ul>                                                                                           |
|  | <ul style="list-style-type: none"><li>• <b>Global trends</b><br/>Analyzed yearly averages of GDP, emissions, and population to identify overall global patterns.</li></ul>                                                                                                   |
|  | <ul style="list-style-type: none"><li>• <b>Overall Highlight</b><br/>The project demonstrates how SQL can combine environmental, economic, and population data to identify country-level and global energy trends, emission patterns, and sustainability insights.</li></ul> |

## Challenges Working

- Understanding table relationships and applying the correct joins
- Ensuring data consistency through foreign key constraints
- Handling aggregation correctly across multiple joined tables
- Extracting time-based trends from date fields stored as VARCHAR

THANK  
YOU

